CS-2008 Numerical Computing **Quiz 1 - BCS-7A** **Fall 2025**

Name:_____ Roll Number: _____

Q1. Given the two points ([a, f(a)], [b, f(b)]), the linear Lagrange polynomial ($f_1(x)$) that passes through these two points is given by

- $f_1(x) = \frac{x-b}{a-b} f(a) + \frac{x-a}{a-b} f(b)$
- $f_1(x) = \frac{x}{b-a} f(a) + \frac{x}{b-a} f(b)$
- $f_1(x) = f(a) + \frac{f(b)-f(a)}{b-a}(x-a)$
- $f_1(x) = \frac{x-b}{a-b} f(a) + \frac{x-a}{b-a} f(b)$

Q2. The following data of the velocity of a body is given as a function of time.

Time (s)	10	15	18	22	24
Velocity (m/s)	22	24	37	25	123

A quadratic Lagrange interpolant is found using three data points, t = 15, 18, and 22. From this information, at what of the times given in seconds is the velocity of the body 26 m/s during the time interval of t = 15 to 22 seconds.

- **20.173**
- **21.858**
- **21.667**
- **22.020**

3. The following data of the velocity of a body is given as a function of time.

Time (s)	0	15	18	22	24
Velocity (m/s)	22	24	37	25	123

If you were going to use quadratic interpolation to find the value of the velocity at t = 14.9 seconds, what three data points of time would you choose for interpolation?

- **0, 15, 18**
- **15, 18, 22**
- **0, 15, 22**
- **0, 18, 24**

4. The following x, y data is given

x	15	18	22
y	24	37	25

The Newton's divided difference second order polynomial for the above data is given by

$$f_2(x) = b_0 + b_1(x - 15) + b_2(x - 15)(x - 22)$$

The value of b_1 is

- **-1.048**
- **0.1433**
- **4.333**
- **24.00**

5. The polynomial that passes through the following x, y data

x	18	22	24
y	?	25	123

is given by

$$8.125x^2 - 324.75x + 3237, \quad 18 \leq x \leq 24$$

The corresponding polynomial using Newton's divided difference polynomial is given by

$$f_2(x) = b_0 + b_1(x - 18) + b_2(x - 18)(x - 22)$$

The value of b_2 is

- **0.2500**
- **8.125**
- **24.00**
- **not obtainable with the information given**

**** Each question is worth of 2 marks.**

For Reference

$$P\ (x) = f(x_0) \cdot \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} \ + \ f(x_1) \cdot \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} \ + \ f(x_2) \cdot \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

$$P(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots$$

Here:

- $b_0 = f[x_0]$
- $b_1 = f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$
- $b_2 = f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$

$$P_2(x) = \sum_{i=0}^{2} f(x_i) \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$$

Rough Workspace

*The rough workspace is intended only for calculations and drafts. Any answers written in the rough work area will not be considered